

99.500 Applied Mathematics for Engineering

Spring Semester AY2026

Instructor: Dr. Hansong Xue
Science, Mathematics and Technology (SMT)

Week 1: Welcome to AME!

- Course description
- Icebreaker: survey & self-introduction
- Non-technical discussion
- Introduction of mathematical modeling in engineering
- About myself:

<https://www.sutd.edu.sg/profile/xue-hansong>

Module Learning Outcomes

At the end of the term, students will be able to:

1. Understand the fundamentals of complex analysis and apply the tools to evaluate integrals and solve complex equations appearing in engineering problems.
2. Understand Laplace and Fourier transforms and apply them to solve differential equations.
3. Understand the fundamentals of ordinary and partial differential equations, be familiar with several important differential equations encountered in engineering problems and know the basic techniques to solve them.

Soft skills enhancement: communication (oral & written), collaboration, critical and creative thinking

More “**applied**” focused (NOT Pure Mathematics/Programming Course)

Icebreaker: survey & self-introduction

<https://forms.office.com/r/sHKUEUs5gk?origin=IprLink>



Icebreaker: survey & self-introduction

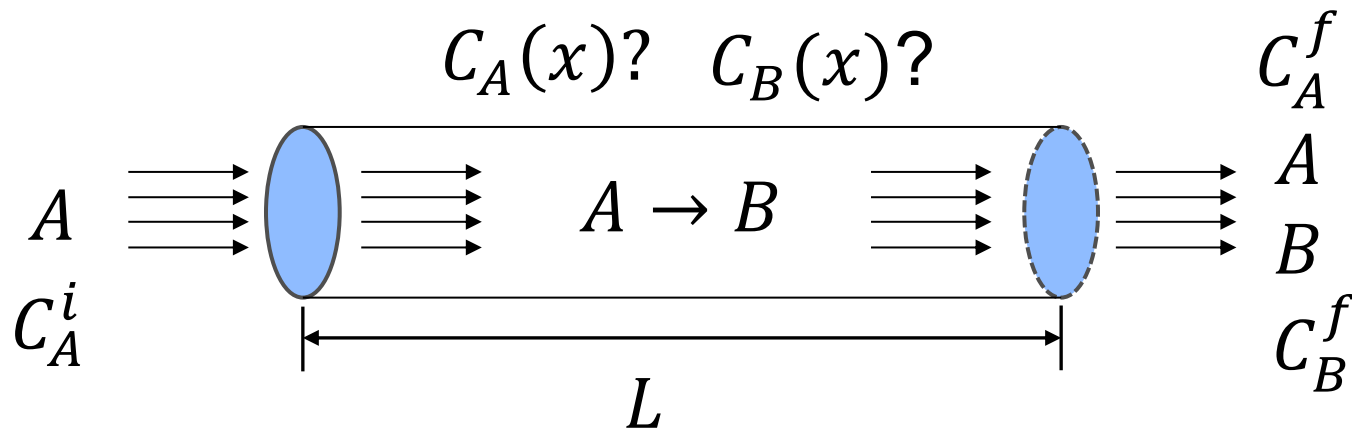
- ✓ Who are you?
- ✓ Which pillar are you from?
- ✓ Why are you here and what do you hope to learn?
- ✓ Share a fun fact about yourself!

Non-technical Discussion

1. What is your expectation in this module?
2. What is instructor expectation in this module?
3. What is the difference between undergraduate and postgraduate study/course?
4. What is the role of student and teacher in learning nowadays?

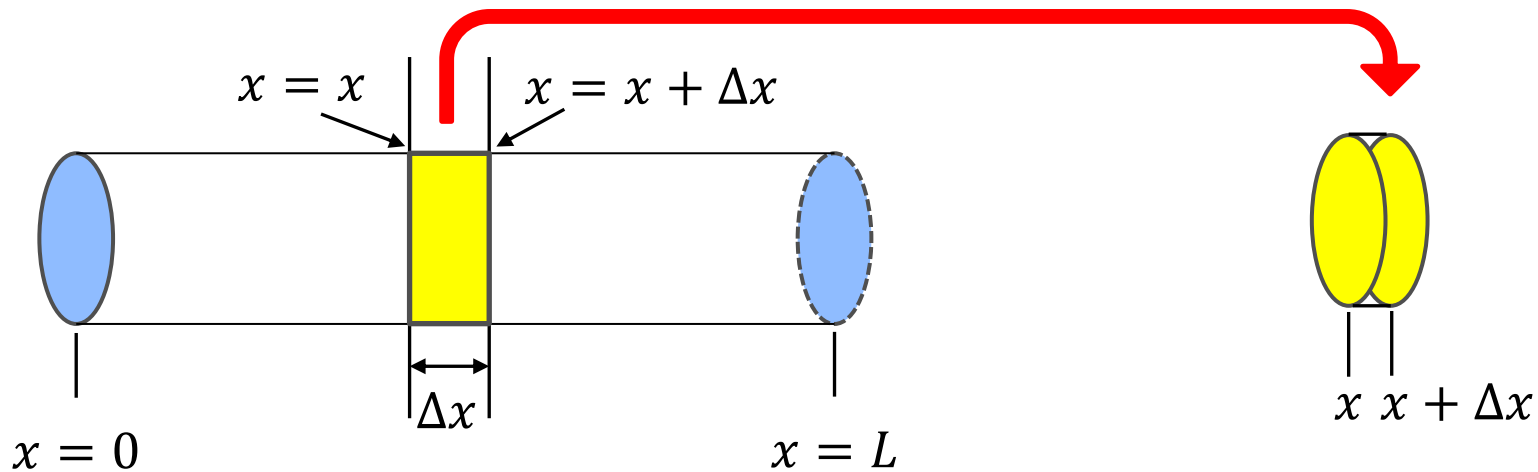
Mathematical Modeling in Flow Reactor

Example 1: Concentration of reactant and product in a flowing tube



1st order reaction: $r_A \rightarrow -kC_A$

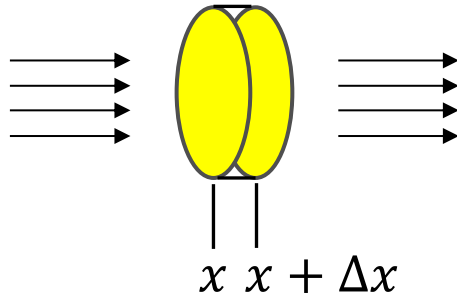
Shell Balance Equation



Shell Balance Equation: for any specific substance, A or B or total of A and B , within any control volume

Accumulation = In – Out + Generation - Consumption

Governing Equation



For reactant A

In: $uC_A|_x A \left(\frac{\text{m}}{\text{s}} \frac{\text{mol}}{\text{m}^3} \text{m}^2 \right)$

Out: $uC_A|_{x+\Delta x} A \left(\frac{\text{m}}{\text{s}} \frac{\text{mol}}{\text{m}^3} \text{m}^2 \right)$

Generation: 0

Rate of reaction: $r_A = -kC_A$

Consumption: $-r_A \Delta V = kC_A A \Delta x$

Accumulation: $\frac{\partial(C_A \Delta V)}{\partial t} = A \Delta x \frac{\partial C_A}{\partial t}$

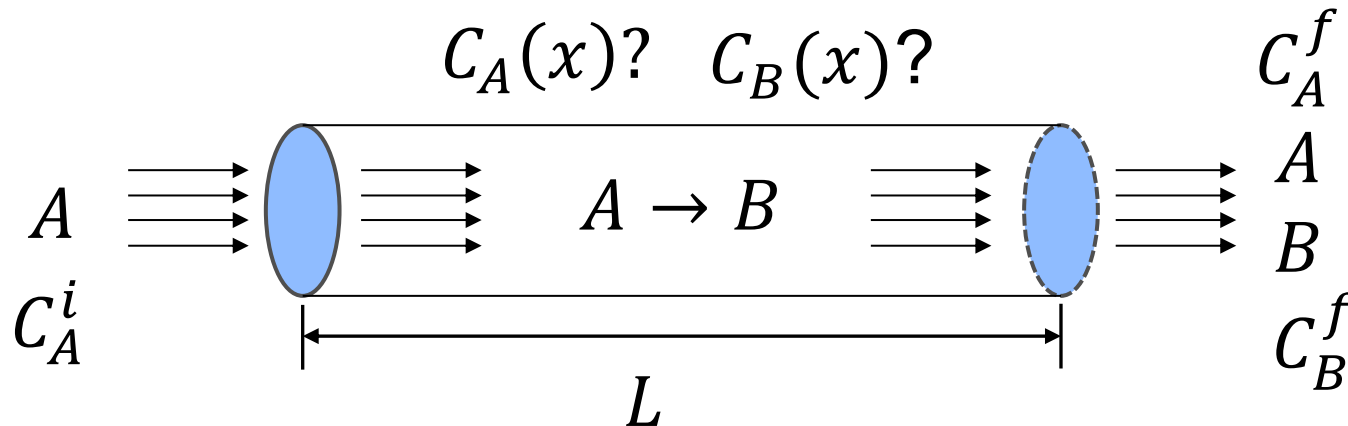
$$A \Delta x \frac{\partial C_A}{\partial t} = uC_A|_x A - uC_A|_{x+\Delta x} A - kC_A A \Delta x$$

$$\frac{\partial C_A}{\partial t} = \frac{uC_A|_x - uC_A|_{x+\Delta x}}{\Delta x} - kC_A$$

$$\frac{\partial C_A}{\partial t} = \frac{-u(C_A|_{x+\Delta x} - C_A|_x)}{\Delta x} - kC_A$$

$$\frac{\partial C_A}{\partial t} = -u \frac{\partial C_A}{\partial x} - kC_A \text{ as } \Delta x \rightarrow 0$$

Constraint Condition



At steady-state: $\frac{\partial C_A}{\partial t} = 0$ & Accumulation is 0

$$0 = -u \frac{dC_A}{dx} - kC_A \quad \text{or} \quad \frac{dC_A}{dx} = -\frac{k}{u}C_A$$

Boundary condition: $C_A(x=0) = C_{A,0} \implies \begin{cases} \frac{dC_A}{dx} = -\frac{k}{u}C_A \\ C_A(x=0) = C_{A,0} \end{cases}$

1st order linear ordinary differential equation (ODE) with initial value

$$\frac{dC_A}{dx} = -\frac{k}{u}C_A = -\frac{dC_B}{dx} \Rightarrow \frac{d(C_A + C_B)}{dx} = 0 \Rightarrow C_A + C_B = \text{const}$$

Assumptions

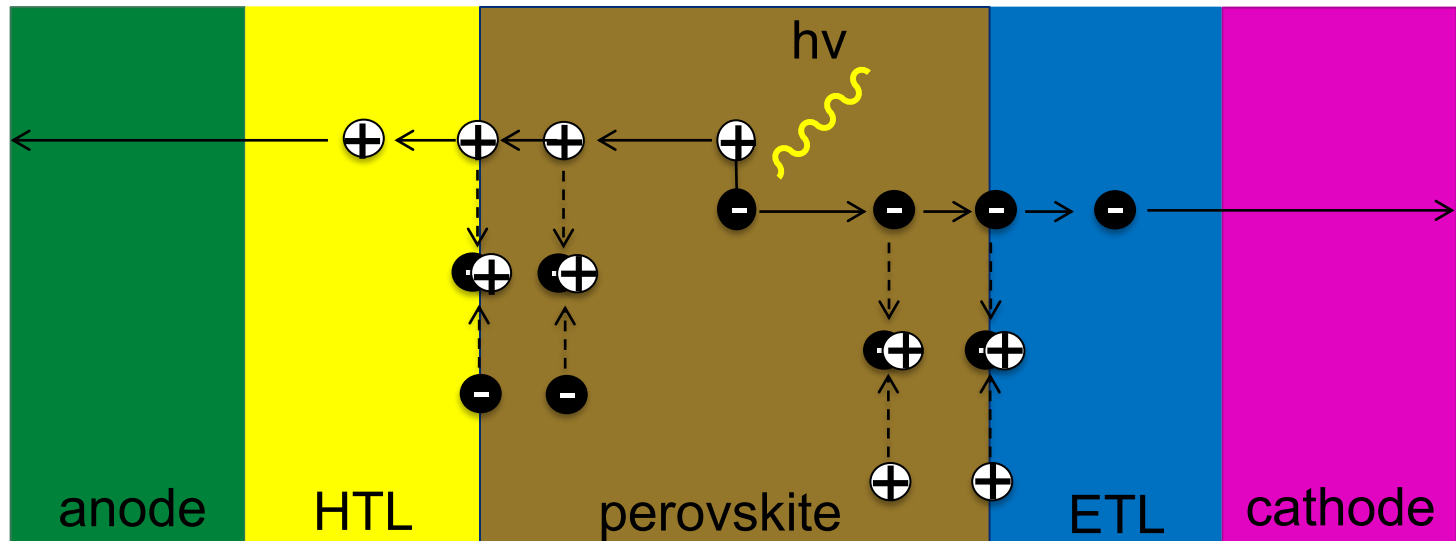
What assumptions did we make to derive this model?

- Ideal flow pattern (uniform flow)
- Constant flow rate
- Constant density (incompressible fluid)
- Isothermal (constant temperature) and rate constant
- No viscous drag force
- Homogeneous 1st order reaction
- No leakage
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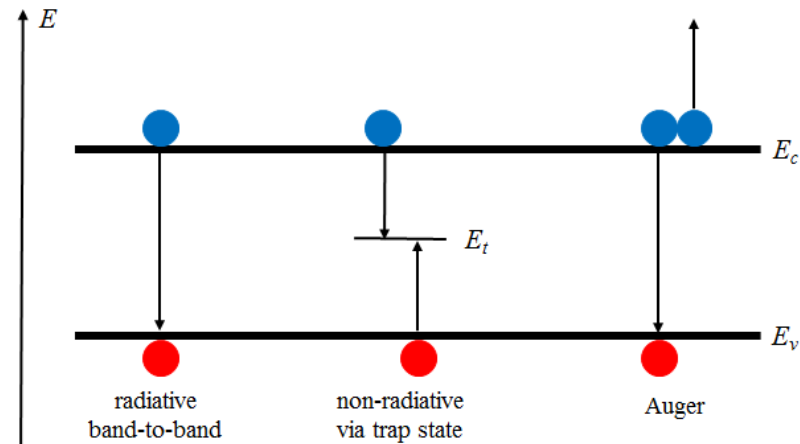
Valid vs Questionable ?

Mathematical Modeling in Solar cell

Example 2: Optoelectronics with electric potential, concentration of electron and hole



1. Charge separation
2. Charge transport
3. Charge recombination



Mathematical Modeling in Solar cell

Example 2: Optoelectronics with electric potential, concentration of electron and hole in perovskite solar cell

$$\begin{aligned}\frac{dj_h(x)}{dx} &= S(x), \\ \frac{dj_e(x)}{dx} &= S(x), \\ \varepsilon \frac{d^2\psi(x)}{dx^2} &= q(c_e(x) - c_h(x)),\end{aligned}\quad \text{Governing equations}$$

h : holes

e : electrons

ε : permittivity

ψ : electric potential

V_0 : potential $x = 0$

μ : charge carrier mobility

q : elementary charge

V_t : thermal voltage

c_i : carrier concentration

j : the charge carrier flux

S : source term for the charge carriers

c_h^0 : hole concentration at $x = 0$

c_e^0 : electron concentration at $x = L$

D : charge carrier diffusion coefficient

g : charge carrier generation rate

c_i : intrinsic carrier concentration

r_{rad} : radiative recombination rate

B : radiative recombination coefficient

Constitutive relations

$$\begin{aligned}j_h(x) &= -\mu_h c_h(x) \frac{d\psi(x)}{dx} - D_h \frac{dc_h(x)}{dx}, \\ j_e(x) &= \mu_e c_e(x) \frac{d\psi(x)}{dx} - D_e \frac{dc_e(x)}{dx},\end{aligned}$$

$$D_{(h,e)} = V_t \mu_{(h,e)},$$

$$S(x) = g(x) - r_{rad},$$

$$r_{rad} = B(c_e c_h - c_i^2),$$

Boundary conditions

$$c_h(0) = c_h^0, j_e(0) = 0, \psi(0) = V_0,$$

$$c_e(L) = c_e^0, j_h(L) = 0, \psi(L) = 0,$$